HaemoSync AI

Strategic Autonomous Pipeline for Emergency Blood Logistics

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01 Problem

- Inventory Silos: Eliminating the lack of visibility between hospital blood banks.
- Manual Coordination: Replacing time-consuming phone calls and manual audits with an autonomous agent.
- High Lead-Time: Reducing the critical delay between identifying a shortage and initiating a transfer

- Decision Latency: Addressing the gap where life-saving blood exists nearby but isn't identified or requested in time.
- Fragmented Data: Turning siloed, unstructured hospital records into a real-time, actionable inventory grid.
- Human Factor Risk: Mitigating the "Single Point of Failure" by automating the redistribution of scarce O-Negative units.

Existing Alternatives

- Traditional Hospital Information Systems (HIS): Static database providers that track inventory but lack autonomous reasoning or cross-hospital orchestration.
- Manual Logistics Coordinators: Human-led dispatch teams that manage transfers via phone and email but are limited by manual auditing speed.
- Third-party Medical Delivery Apps: Delivery platforms that focus on the physical transport but do not have the logic to autonomously identify the optimal supplier-recipient match.

- Manual Communication Loops: Relying on fragmented phone calls, WhatsApp groups, and emails between blood bank administrators to find stock.
- Siloed Inventory Sheets: Hospital-specific spreadsheets or paper records that are not shared in real-time across a regional grid.
- Reactive Procurement: Waiting for a critical shortage to occur before manually searching for units, leading to high lead-time and increased patient risk.

02 Solution

- Autonomous Monitoring: We provide a system that continuously scans hospital inventory data, removing the burden of manual auditing from staff.

- Intelligent Triage: By using AI reasoning, the system automatically identifies which hospital needs blood and which can supply it based on real-time data.

- Instant Alerts: We bridge the communication gap by sending immediate, structured emergency dispatches directly to the decision-makers' inboxes

- Seamless Logistics: Customers experience a shift from fragmented, siloed operations to a fully "connected hospital grid".
- Reliability & Speed: The platform provides a "zero-latency" response where transfers are initiated the moment a shortage threshold is hit.
- Data-Driven Confidence: Administrators gain peace of mind knowing that resource allocation is governed by optimized algorithms rather than guesswork

04 Unique Value Proposition

- Autonomous Agentic Reasoning: Unlike standard dashboards, the system uses an AI agent (Gemini 2.5 Flash) to analyze inventory and autonomously decide on the best logistics move.
- Operations Research Integration: The system applies heuristic-based greedy algorithms to perform inter-hospital transshipment in real-time.
- Modular Automation: Built on n8n, it creates a seamless "Data-to-Reasoning-to-Action" pipeline that triggers emergency dispatches without human intervention.

- Zero-Latency Triage: Traditional solutions rely on manual coordination (phone calls/emails), whereas HaemoSync identifies and alerts for transfers instantly.
- Network-Level Optimization: Most systems are siloed to one hospital; this creates a transparent, connected grid that optimizes existing stock across an entire region.
- Scalable & Sovereign: The architecture is designed to eventually run on local, open-source LLMs to ensure 100% medical data privacy while remaining highly cost-effective.

05 Unfair Advantage

- Operational Heuristic Integration: Unlike standard AI implementations, we have integrated specific Operations Research (OR) heuristics, such as the greedy algorithm, directly into our agentic reasoning layer to solve complex inter-hospital transshipment problems.
- Zero-Human-Latency Orchestration: Our custom n8n workflow creates a unique "Data-to-Reasoning-to-Action" pipeline that eliminates the manual delays inherent in traditional emergency communication loops.
- Hybrid Intelligence Framework: We combine the high-level reasoning of Google Gemini 2.5 Flash with the structural precision of centralized medical inventory databases to provide optimized, life-saving decisions at the speed of an emergency.

- Specialized Logistics Logic: Our unique logic for identifying the most critical recipient and the most capable supplier based on real-time inventory thresholds.
- Modular Architecture: A highly adaptable and scalable system designed to eventually incorporate IoT triggers and local, sovereign AI models for enhanced data privacy.
- Operations Research Foundation: A foundational approach built on core engineering principles that views healthcare logistics as a network optimization problem

07 Customer Segments

- Hospital Administrators: Staff responsible for logistical oversight and emergency response coordination.
- Blood Bank Technicians: Personnel who manage daily inventory and need immediate alerts for critical shortages.
- Emergency Response Teams: Medical staff who require rapid confirmation of blood availability for life-saving procedures

- Regional Hospital Networks: Large healthcare groups (like the Thiruvananthapuram medical corridor) looking to optimize resource sharing.
- Government Health Departments: Entities like the Kerala Health Ministry aiming to build a more resilient state-wide blood grid.
- Private Healthcare Chains: Organizations that want to automate logistics to reduce operational waste and improve patient outcomes.

06 Channels

- Direct Government Partnerships: Collaborative pilots with the Kerala Health Ministry and regional medical councils to integrate with state-wide blood grid initiatives.
- Medical HIS Integration: Partnering with Hospital Information System (HIS) providers to embed HaemoSync AI directly into existing administrative dashboards.
- Professional Networks: Utilizing LinkedIn and academic conferences to showcase the prototype's reasoning capabilities to hospital administrators and Operations Research professionals.
- Strategic Emergency Corridors: Direct outreach to hospitals within the Thiruvananthapuram medical network to establish the initial "connected grid".

Early Adopters

- Regional Blood Bank Technicians: Personnel at major hubs like the Government Medical College who manage daily O-Negative stock levels.
- Emergency Dispatch Coordinators: Staff responsible for orchestrating the physical transfer of units between facilities.
- Logistics Beta-Testers: Early-stage administrators in the Thiruvananthapuram medical network willing to pilot automated SMTP alerts

- Critical Patients: Individuals in life-threatening situations requiring immediate O-Negative blood transfusions.
- Hospital Administrators: Decision-makers who gain a reduction in "Lead-Time" and a more resilient, connected hospital grid.
- The Healthcare System: Regional medical councils that benefit from optimized resource allocation and a reduction in manual coordination errors.

03 Key Metrics

- TTT (Time-to-Transfer) Monitoring: Tracking the reduction in minutes from the moment a shortage is identified in the Google Sheets database to the final Gmail dispatch.
- System Latency Audits: Measuring the processing speed of the n8n orchestration layer and Gemini reasoning engine to ensure near-instantaneous triage.
- Data Accuracy Logs: Reviewing the consistency of inventory data pulled from the hospital network into the centralized repository

- Rebalancing Accuracy: Achieving a high percentage of correct supplier-to-recipient matches based on the greedy algorithm's optimization logic.
- Zero-Delay Dispatch: Successful delivery of structured, high-priority transfer alerts to administrators for 100% of identified critical shortages.
- Resource Utilization Rate: Maximizing the redistribution of existing stock to eliminate preventable "stock-out" situations across the grid

08 Cost Structure

- Serverless n8n Hosting: Minimal monthly costs for maintaining the orchestration layer on a stable serverless environment.
- System Maintenance: Occasional costs for database management and ensuring the Google Sheets inventory bridge remains secure.
- Domain & Security: Small fixed annual costs for secure API endpoints and professional communication domains

- API Usage-Based Pricing: Costs for the Google Gemini Reasoning Engine that only scale based on the number of emergency inventory checks performed.
- Data Transfer Fees: Minor costs associated with pulling inventory data and pushing automated SMTP (Gmail) alerts.
- Open-Source LLM Scaling: If transitioning to local models (like Llama 3), variable costs would include specialized GPU compute hours during peak demand

09 Revenue Streams

- Operational Efficiency Gains: Monetizing the significant reduction in man-hours previously spent on manual coordination and inventory auditing.
- Inventory Loss Mitigation: Generating value by drastically reducing the number of expired blood units through optimized, timely redistribution.
- System Integration Fees: Fees associated with connecting the autonomous agentic pipeline to existing hospital databases

- Public Health Sector (Government): Value-based contracts focused on lowering "Total System Safety Stock" and improving public emergency response times.
- Private Hospital Chains: A SaaS (Software as a Service) Model providing a subscription-based "Autonomous Logistics Layer" for multi-branch coordination.
- Regional Medical Councils: Performance-based incentives for hitting specific "Time-to-Transfer" (TTT) benchmarks across the connected grid

Future?

IOT Integration- Implementing emergency panic buttons in ERs and automated refrigerator threshold triggers for real time shock alerts

Sovereign AI Deployment – Transitioning to local open-source LLMs like Llama 3 to ensure 100% medical data privacy and security

Geospatial Optimization – Utilizing Google Maps API for shortest path routing with real-time traffic data for blood delivery couriers

Proof Of Concept

The Sample Data

HaemoSync_Inventory

File Edit View Insert Format Data Tools Gemini Extensions Help

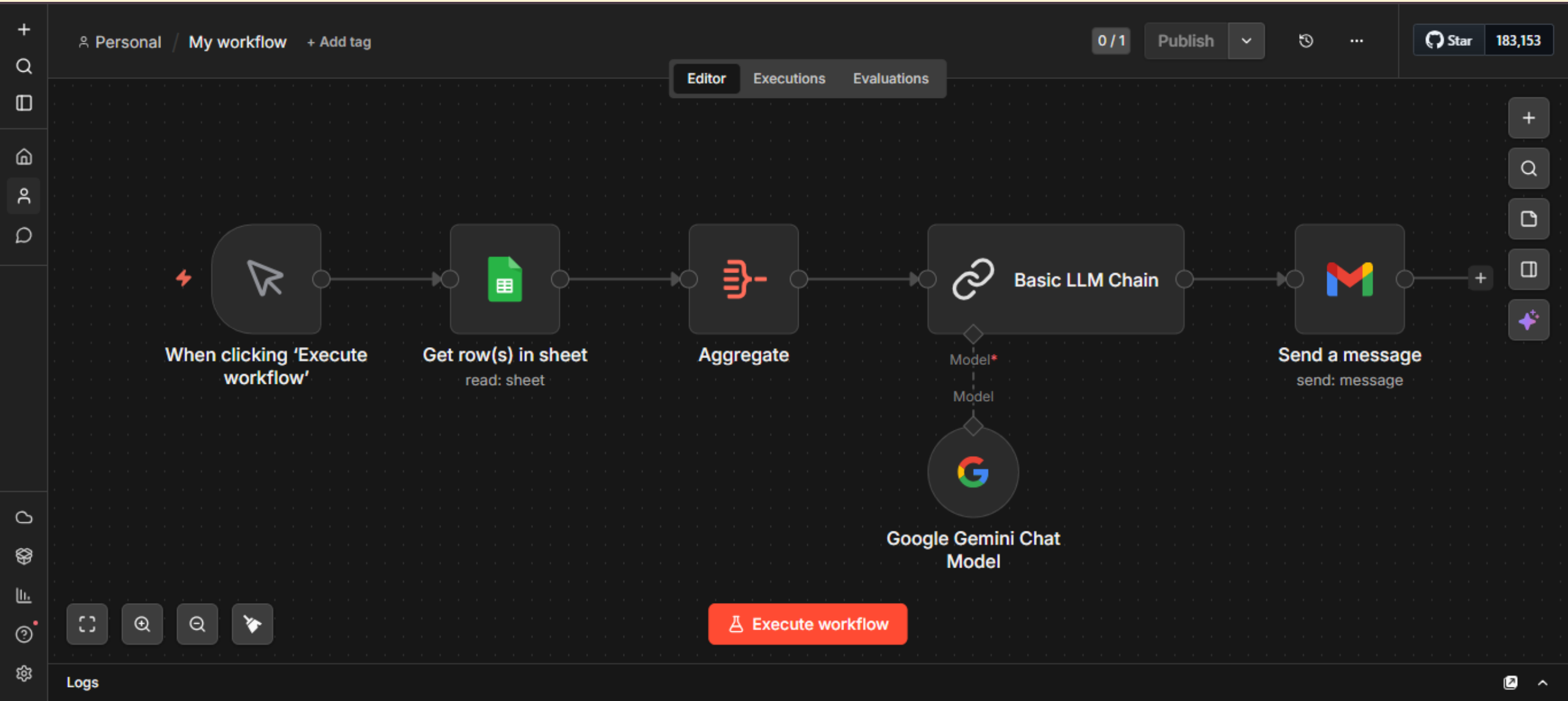
Share

G8

Summarise this data

	A	B	C	D	E	F	G	H	I	J
1	Hospital Name	O Negative Units	A Positive Units	Emergency Status	Contact Email					
2	KIMS Hospital	1	20	CRITICAL	akshayviswanathsarma@gmail.com					
3	Medical College	18	15	NORMAL	akshayviswanathsarma@gmail.com					
4	SP Fort Hospital	8	12	NORMAL	akshayviswanathsarma@gmail.com					
5	PRS Hospital	10	5	NORMAL	akshayviswanathsarma@gmail.com					
6	Ananthapuri Hospital	7	25	NORMAL	akshayviswanathsarma@gmail.com					
7										

The Workflow



The Output

🚨 HaemoSync AI: Emergency Blood Transfer Alert Inbox x

akshayviswanathsarma@gmail.com 00:58 (14 hours ago)

to me

Subject: URGENT: Critical O-Negative Blood Shortage - Immediate Transfer Required Dear Hospital Administrator, This is an urgent notification from HaemoSync AI regarding a critical shortage of O-Negative blood units within the Thiruvananthapuram network. Our real-time inventory analysis indicates that KIMS Hospital, Thiruvananthapuram currently has only 1 unit of O-Negative blood, representing a life-threatening shortage for immediate patient needs. Concurrently, Ananthapuri Hospital and Research Institute has a surplus of 10 O-Negative units available for transfer. To mitigate this critical risk, immediate action is required. ACTION REQUIRED: Transfer 5 units of O-Negative from Ananthapuri Hospital and Research Institute to KIMS Hospital, Thiruvananthapuram immediately. Please confirm receipt of this directive and the commencement of the transfer process as soon as possible. Your prompt cooperation is vital in ensuring patient safety across the region. Sincerely, HaemoSync AI Autonomous Emergency Response Agent Blood Bank Logistics, Thiruvananthapuram

Demonstration
A video has been attached to the Project Zip File

The current PoC successfully demonstrates the autonomous dispatch pipeline. Future iterations will refine the greedy algorithm's directionality to ensure supply always flows from high-surplus to high-deficit nodes

Since The PoC uses an API Key , the A.I agent may hallucinate at times and produce irrelevant results
But the integration of local LLM’s into Hospital Databases can solve the problem and be helpful statewide

Thank You!